

STUDENT NUMBER: 2805001611

19. API is a set of rules and protocols that allow different software programs to communicate with each other

- APIs are important because they act as secure, standardised intermediaries that allow different systems to communicate, share data and utilise functionality without direct integration

20. An API key is a unique alphanumeric string used to identify and authenticate an application or project making a request to an API.

- An API key purpose is to serve as an intermediary that allows two different software programs to communicate and exchange data with each other.

How it should be stored	If exposed publicly
- Use environment variables	- Financial loss
- Apply the principle of least privilege	- Legal and Regulatory Penalties
- Avoid Client side exposure	- Service disruption

21. A REST API is an architectural style for designing networked applications that allows different systems to communicate over the internet.

- REST stands for Representational State Transfer.

* GET - used to retrieve a representation of a resource.

* POST - Used to submit data to a server to create a new subordinate resource.

* PUT - Used to replace an entire existing resource with new data or create a resource at a specific URL if it doesn't exist

* DELETE - used to permanently remove a specific resource from the server

22. JSON is a light weight, text based data interchange format used to store and transmit structured data primarily between a server and a web application.

- JSON is commonly used when working with API and data exchange because it does not use bulky closing tags.

Example of a JSON object

- "First Name": Vutlhari
- "Last Name": Kubayi

23. Prompt Engineering is the process of designing, refining and optimising inputs (prompts) to guide generative AI models towards producing accurate, relevant and high quality outputs.

- It is an important skill because it transforms AI tools from generic, conversational bots into precise high-performance thinking partners.

Examples

- Event foster
- Restaurant menu

24. A MAP function is a high level tool used in programming, spreadsheet and mathematics to transform data.

- In programming and data processing MAP function utilise code, algorithms and logical structures to automate these transformations, converting unorganised inputs into structured outputs like reports, graphs or actionable insights.

Example in Java script

```
* Console.log ("Hello, world!")
```

25. Cloud Computing is the on-demand delivery of IT resources such as computing power, storage and data bases over the internet with pay-as-you-go pricing.

26 Version Control System is software that tracks and manages changes to files - primarily source code - over time, creating a recorded history of modifications.

Collaboration without Conflicts: multiple analysts can work on the same projects simultaneously. Git's merging capabilities combine their different contributions while identifying exactly where changes might conflict.

27 Jupyter Notebook is an open-source web-based application designed for creating and sharing interactive computational documents. It allows users to combine live code, visualisations and narrative text in a single document, making it highly useful for data analysis, scientific computing and machine learning workflows.

- It is mostly used with the Python programming language.
- The popularity of Jupyter Notebooks among data analysts and data scientists stems from its ability to facilitate interactive and iterative data exploration, visualisation and reporting.

28. Python is a high-level, interpreted, and object-oriented programming language known for its simplicity, readability and versatility. It emphasises clear syntax and dynamic semantics making it accessible for both beginners and experienced developers.

- Its widespread use in data analysis, artificial intelligence (AI), machine learning, web development, automation and scientific computing explains why it is considered one of the most popular languages for data analysts and AI.

Four Libraries commonly used

- NumPy: Provides support for large multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these data structures.
- Pandas: Offers data structures and data analysis tools, making it easier to manipulate structured data, such as tables and time series.
- Matplotlib: A plotting library used for creating static, animated and interactive visualisation in Python